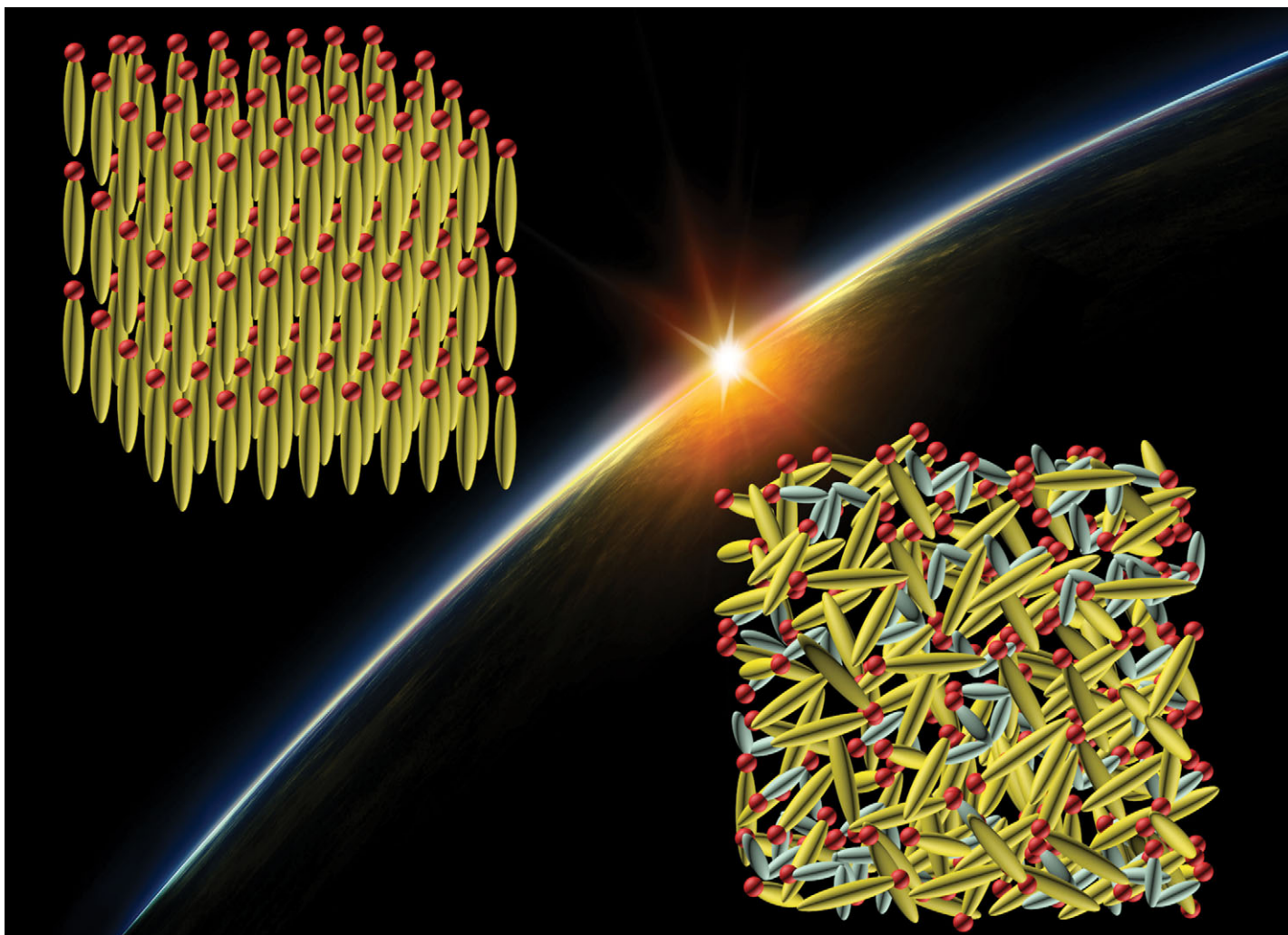


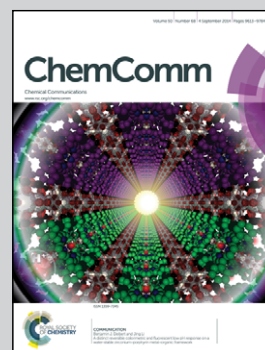


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Photoresponsive liquid crystals based on halogen bonding of azopyridines

A series of halogen-bonded LCs were constructed using molecular halogen and azopyridine, which show photoinduced phase transition upon UV irradiation, indicating their photonic applications as other photoresponsive LC materials.

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Photoresponsive liquid crystals based on halogen bonding of azopyridines†

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A series of photoresponsive halogen-bonded liquid crystals (LCs) were successfully constructed using molecular halogen and azopyridine compounds, which show interesting properties of photoinduced phase transition upon UV irradiation. In addition, bromine-bonded LCs were first obtained with high mesophase stability.

As the fourth state of matter, liquid crystals (LCs) show both the controllable fluidity of an isotropic liquid and the ordered regularity of a crystalline solid, and greatly influence our daily life.¹ Azobenzene (AZ) is one of the typical and widely studied photochromic molecules, exhibiting reversible *trans/cis* isomerization upon photoirradiation.^{1,2} Combining the photoresponsive properties of AZs with the self-organization feature of LCs, photocontrollable LC actuators have been fabricated which show potential for a variety of applications in diverse fields, such as flat panel displays,^{1a} photonics,^{1b} photo-driven devices,^{1c,d} and more recently in nanotechnology.^{1e}

Azopyridine (AzPy) derivatives have the advantage of possessing both the light activity related to AZs and the capability of self-assembly through pyridyl groups. For example, they can form hydrogen bonds with organic acids,³ ionic bonds with inorganic acids,⁴ organic ligands with metal ions,⁵ and halogen bonds with halides.⁶ AzPy derivatives provide a chance for further functionalization of photoresponsive LCs *via* different versatile approaches, such as hydrogen bonding (HB) or halogen bonding (XB), instead of the use of covalent interactions.⁷

A typical halogen bond can be described in general as D...X-Y, where X is the electrophilic halogen atom (Lewis acid, XB donor),

D is the donor of electron density (Lewis base, XB acceptor), and Y is a carbon, nitrogen or halogen atom.⁸ The binding nature of XB results in the D...X distances being shorter than the sum of the van der Waals radii of the involved atoms, and the stronger the XB the shorter the D...X distance.⁹ Consistent with the rationalization of XB being an electron donation from D to the anti-bonding X-Y orbital, the formation of XB slightly lengthens the covalent X-Y bond.¹⁰ More importantly, the two noncovalent interactions of XB and HB exhibit a little difference. Firstly, XB is more directional than HB. The angle between the covalent and noncovalent bonds around the halogen atom in D...X-Y is approximately 180°, so that the higher directionality of XB with respect to HB provides a more rigid junction. Secondly, its interaction strength can be tuned by choosing suitable halogen atoms that take part in the bond formation without significantly changing the electronic structure of the compound.¹¹ With the exception of these differences, most of the energetic and geometric trends, as well as other features observed by spectroscopic and theoretical investigations, have also been found with XB.^{10,12}

In 2004, Bruce *et al.* demonstrated the use of XB as a directing force in LC formation and studied interactions between 4-alkoxy-stilbazoles and iodopentafluorobenzenes.^{13a} Then he reported LCs formed with XB, namely complexes of molecular iodine with alkoxy-stilbazoles.^{13b} This system had high mesophase stability, which was predicated on the intermolecular I...I contact. Recently, Priimagi *et al.* demonstrated that the higher directionality of XB provides a more rigid dye-polymer junction, resulting in enhanced surface-relief grating.¹¹ Meanwhile, Marco Saccone *et al.* studied XB donor molecules for assembling photosensitive co-crystals.¹⁴

The above-mentioned features of XB are shown to provide unique possibility in understanding and controlling the properties of soft matter. In the present work, we designed a series of supramolecular LC complexes using halogen molecules and AzPy compounds to study the role of both the nature and the strength of XB in LC formation and photoresponsive properties. As shown in Fig. 1, halogen-bonded complexes *n*X were prepared using halogen molecules and AzPy derivatives having different

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† Electronic supplementary information (ESI) available: The molecular structures, synthesis, characterization and photoresponse of compounds. The videos of preparation of halogen-bonded complexes and photoinduced phase transition behaviours of the halogen-bonded complexes. See DOI: 10.1039/c4cc02344g

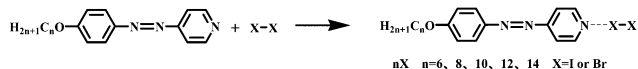
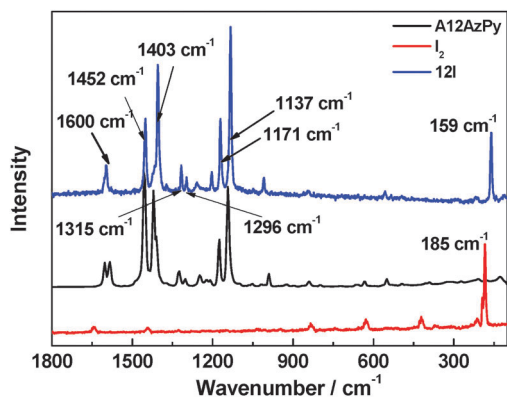


Fig. 1 Possible molecular scheme of halogen complexes of AnAzPy.

Fig. 2 Raman spectra of I₂, A12AzPy and 12I.

alkyl chains (AnAzPy). Detailed information of synthesis, characterization, measurements and instrumentation are described in the ESI†.

Raman spectroscopy is a powerful method for characterization of XB, as the halogen-stretching frequency is Raman active in both the complexed and uncomplexed molecules.¹⁵ As shown in Fig. 2, the I–I stretching peak in molecular iodine was observed at 185 cm^{−1}. But this typical peak in the iodine-bonded complex disappeared and a new band at 159 cm^{−1} appeared, which can be ascribed to higher electron-donating ability of AzPy derivatives. This novel absorption in Raman spectroscopy can be with certainty assigned to the stretching of the donor-halogen charge-transfer bond of the XB interaction.¹⁶ This also suggests that iodine molecules interact with AzPy derivatives, proving the formation of XB *via* N⋯I interactions. Meanwhile, the N=N stretching gives rise to a band at 1403 cm^{−1}, and the band at 1600 cm^{−1} was ascribed to the benzene quadrant stretching. The band at 1452 cm^{−1} was attributed to AZ ring vibration. Bands between 1315 and 1296 cm^{−1} can be ascribed to aromatic ring vibrations, which overlap with the C–H deformations of aliphatic side-chains.¹⁷ The Raman spectra of Br₂, A12AzPy and 12Br are given in Fig. S3 (ESI†). The formation of XB was further confirmed by X-ray photoelectron spectroscopy (XPS), as shown in Fig. S4 and S5 (ESI†).

The LC properties of halogen-bonded complexes were measured by using polarizing optical microscopy (POM) and differential scanning calorimetry (DSC), which are summarized in Table 1.

As shown in Table 1, the thermal properties of 10I, 12I and 14I are similar, which directly melted to mesophases and then reached the clearing point upon heating. All of them showed focal conic fan textures as observed by POM at their LC temperatures, as shown in Fig. 3a and Fig. S6 (ESI†). These LC textures often appear in smectic LCs. To further identify these mesophases, both the SAXS experiments (Fig. S7, ESI†) and LC cells with homeotropic alignment (Fig. S8, ESI†) were performed, indicating the smectic A phase of the obtained halogen-bonded complexes. However, no LC behaviours were observed for 6I and 8I.

To evaluate the bonding capability of AzPy derivatives with different halogens, molecular bromine (Br₂) was chosen as a Lewis base to prepare bromine-bonded complexes. Having found iodine complexes, Bruce attempted to prepare bromine analogues by allowing interaction of an alkoxystilbazole with bromopentafluorobenzene and molecular bromine, respectively. Despite several attempts, they never obtained the bromine-bonded LCs, and it is argued that the strength of halogen bonds decreases in the following order: I > Br > Cl (*e.g.* N⋯Br interactions tend to be > 3 Å in the solid state),^{13,16} Interestingly, we first prepared bromine-bonded LC compounds with molecular bromine, as shown in Table 1 and Fig. S9 and S10 (ESI†). Moreover, the bromine-bonded supramolecular LCs showed high mesophase stability in this system, as shown in Fig. S11 (ESI†). Hence, the results of the present AzPy–halogen system suggest that the N⋯Br interaction is strong enough to form ordered mesophases.

Then the photoresponsive properties of the halogen-bonded samples in their LC phases were studied upon UV irradiation at 360 nm,¹⁸ which were *in situ* observed by POM. As shown in Fig. 4, the initial phase was smectic A at about 120 °C before photoirradiation, where the 12I should be in the *trans* form. Upon UV irradiation, the birefringent LC texture disappeared quickly and only a dark image was obtained, indicating the occurrence of photoinduced phase transition from an LC to an isotropic phase (Video S2, ESI†). Obviously, the phase transition is the result of photoisomerization of AzPy molecules from their *trans* to *cis* form as the bent-shaped *cis* isomers tend to destabilize the ordered LC phase.³ The initial LC texture was recovered after the isotropic sample was irradiated with visible light for 80 s. Similarly to other AZ-containing LC materials,^{1–4} this photochemical phase transition is brought about by the reversible photoisomerization of AzPy units.^{3,18} In this photoresponsive process, a large change in birefringence can be reversibly induced because of the LC-to-isotropic phase transition,

Table 1 Summary of thermal properties of halogen-bonded compounds^a

Materials	<i>T</i> /°C	Materials	<i>T</i> /°C	Materials	<i>T</i> /°C
I ₂	C 113.5 I	6I	—	6Br	C 75.1 SmA 108.8 I
A6AzPy	C 56.7 I	8I	—	8Br	C 100.3 SmA 116.0 I
A8AzPy	C 67.9 I	10I	C 123.6 SmA 135.5 I	10Br	C 100.3 SmA 107.5 I
A10AzPy	C 65.9 I	12I	C 119.6 SmA 135.0 I	12Br	C 70.5 SmA 158.3 I
A12AzPy	C 72.8 I	14I	C 117.2 SmA 133.8 I	14Br	C 94.8 SmA 107.3 I
A14AzPy	C 73.3 I				

^a C: crystal; I: isotropic phase; SmA: smectic A phase.

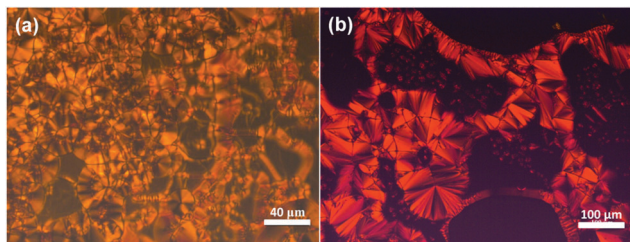


Fig. 3 POM pictures of 12I at 90 °C (a) and 6Br at 93 °C (b) upon cooling from the isotropic phase.

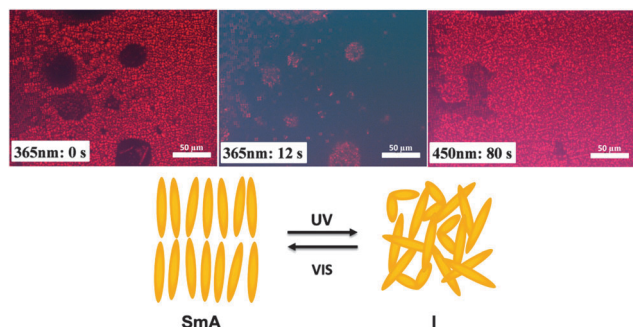


Fig. 4 POM observation of 12I at its LC phase upon UV irradiation. The right picture was obtained after irradiation with visible light for 80 s.

which has potential applications similar as that of the other photoresponsive LC materials.^{1–4,18} However, compared to iodinated LCs, we have not observed the photochemical phase transition in brominated compounds induced by UV irradiation.

In conclusion, supramolecularly halogen-bonded LCs were successfully fabricated using AzPy derivatives with molecular iodine or bromine. The existence of XB interactions was confirmed by XPS, Raman spectroscopy and SAXS results. Meanwhile, the supramolecular LC compounds with bromine were first prepared with high stability, indicating that the N $\cdots$ Br interaction is strong enough to form ordered mesophases in this system. The iodine-bonded LCs showed reversibly photoinduced phase transition upon UV irradiation, just like other photoresponsive LC materials containing covalently or hydrogen bonded AZs. However, we have not observed the photochemical phase transition in brominated compounds induced by UV irradiation. This study would provide valuable knowledge of XB interactions for designing and manipulating novel supramolecularly functional materials.

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Notes and references

- (a) D. Demus, J. Goodby, G. W. Gray, H.-W. Spiess and V. Vill, *Handbook of Liquid Crystals*, Wiley-VCH, Weinheim, Germany, 1998; (b) Q. Li, *Liquid Crystals Beyond Displays: Chemistry, Physics, and Applications*, John Wiley & Sons, 2012; (c) H. F. Yu and T. Ikeda, *Adv. Mater.*, 2011, **23**, 2149; (d) O. S. Bushuyev, A. Tomberg, T. Friščić and C. J. Barrett, *J. Am. Chem. Soc.*, 2013, **34**, 12556–12559; (e) H. F. Yu, *Prog. Polym. Sci.*, 2014, **39**, 781.
- (a) T. Seki, S. Nagano and M. Hara, *Polymer*, 2013, **54**, 6053; (b) T. Seki, *Macromol. Rapid Commun.*, 2014, **35**, 271; (c) A. Goulet-Hanssens and C. Barrett, *J. Polym. Sci., Part A*, 2013, **51**, 3058.
- (a) L. Cui and Y. Zhao, *Chem. Mater.*, 2004, **16**, 2076; (b) D. C. Rogness, P. J. Riedel and K. N. Wiegel, *Liq. Cryst.*, 2006, **33**, 567; (c) L. Cui, S. Dahmane and Y. Zhao, *Macromolecules*, 2005, **38**, 2076; (d) V. A. Mallia, P. K. S. Antharjanam and S. Das, *Liq. Cryst.*, 2003, **30**, 135; (e) H. Liu, T. Kobayashi and H. F. Yu, *Macromol. Rapid Commun.*, 2011, **32**, 378; (f) H. F. Yu, H. Liu and T. Kobayashi, *ACS Appl. Mater. Interfaces*, 2011, **3**, 1333.
- (a) K. Binnemans, *Chem. Rev.*, 2005, **105**, 4148; (b) N. Iranpoor, H. Firouzabadi, D. Khalili and S. Motevali, *J. Org. Chem.*, 2008, **73**, 4882; (c) W. Zhou and H. F. Yu, *RSC Adv.*, 2013, **3**, 22155; (d) W. Zhou and H. F. Yu, *ACS Appl. Mater. Interfaces*, 2012, **4**, 2154; (e) W. Zhou, T. Kobayashi, H. Zhu and H. F. Yu, *Chem. Commun.*, 2011, **47**, 12768.
- (a) P. K. Sudhadevi Antharjanam, V. A. Mallia and S. Das, *Chem. Mater.*, 2002, **14**, 2687; (b) S. Goswami, A. R. Chakravarty and A. Chakravorty, *Inorg. Chem.*, 1981, **20**, 2246.
- (a) C. P. Andrieux, J.-M. Savéant and K. B. Su, *J. Phys. Chem.*, 1986, **90**, 3815; (b) J. Bertran, L. Gallardo, M. Moreno and J. M. Saveant, *J. Am. Chem. Soc.*, 1992, **114**, 9576.
- T. Takashi and J. Fréchet, *Liq. Cryst.*, 2006, **33**, 1429.
- (a) P. Metrangolo, H. Neukirch, T. Pilati and G. Resnati, *Acc. Chem. Res.*, 2005, **38**, 386; (b) E. Corradi, S. V. Meille, P. Metrangolo and G. Resnati, *Angew. Chem.*, 2000, **112**, 1852.
- (a) J. S. Murray, P. Lane, T. Clark and P. Politzer, *J. Mol. Model.*, 2007, **13**, 1033; (b) A. Karpfen, *Struct. Bonding*, 2008, **126**, 1.
- P. Metrangolo, F. Meyer, T. Pilati and G. Resnati, *Angew. Chem., Int. Ed.*, 2008, **47**, 6114.
- (a) A. Priimagi, G. Cavallo and A. Forni, *Adv. Funct. Mater.*, 2012, **22**, 2572; (b) A. Priimagi, G. Cavallo, P. Metrangolo and G. Resnati, *Acc. Chem. Res.*, 2013, **46**, 2686; (c) A. Priimagi, M. Saccone, G. Cavallo, A. Shishido, T. Pilati, P. Metrangolo and G. Resnati, *Adv. Mater.*, 2012, **24**, OP345–OP352.
- (a) P. Metrangolo, G. Resnati, T. Pilati, R. Liantonio and F. Meyer, *J. Polym. Sci., Part A*, 2007, **45**, 1; (b) I. Alkorta, L. Rozas and J. Elguero, *J. Phys. Chem. A*, 1998, **102**, 9278.
- (a) H. L. Nguyen, P. N. Horton, M. B. Hursthouse, A. C. Legon and D. W. Bruce, *J. Am. Chem. Soc.*, 2004, **126**, 16; (b) L. J. McAllister, C. Präsang, J. P.-W. Wong, R. J. Thatcher, A. C. Whitwood, B. Donnio, P. O'Brien, P. B. Karadakov and D. W. Bruce, *Chem. Commun.*, 2013, **49**, 3946.
- M. Saccone, G. Terraneo, T. Pilati, G. Cavallo, A. Priimagi, P. Metrangolo and G. Resnati, *Acta Crystallogr., Sect. B*, 2014, **B70**, 149–156.
- (a) M. Erdélyi, *Chem. Soc. Rev.*, 2012, **41**, 3547; (b) X. Pang, H. Wang, X. R. Zhao and W. J. Jin, *Dalton Trans.*, 2013, **42**, 8788.
- (a) H. Stammreich, R. Forneris and Y. Tavares, *Spectrochim. Acta*, 1961, **17**, 1173; (b) P. Klæboe, *J. Am. Chem. Soc.*, 1967, **89**, 3667.
- P. Metrangolo and G. Resnati, *Chem. – Eur. J.*, 2001, **12**, 2511.
- (a) T. Ikeda, *J. Mater. Chem.*, 2003, **13**, 2037–2057; (b) H. F. Yu, *J. Mater. Chem. C*, 2014, **2**, 3047.